

Radical Equations and Extraneous Roots

Answer Key

Algebra 2

Quick Answers

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| 1. 11 | 3, — |
| 2. 36 m | 7. 4 |
| 3. (a) the solution = 3, (b) left side at $x = -1 = 1$ | 8. (a) the lookout height = 100 m, (b) sighting distance from 250 m = 56.45 km |
| 4. 17 | 9. (a) the solution = 5, (b) left side at $x = 0 = 4$ |
| 5. 84 ft | 10. (a) the solution = 4, (b) the check in the original equation = 1 |
| 6. (a) the solution = 9, (b) left side at $x = 2 =$ | |
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What is verified

15 of 16 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problem 6. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2

HSA-REI.A.2 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–5 of 5 across 16 tagged checks.

Common wrong answers

3. *If they got 3, -1: kept the extra candidate the squaring step produced and reported both $x = 3$ and $x = -1$*
6. *If they got 9, 2: reported both candidates from the squared equation, $x = 9$ and $x = 2$*
9. *If they got 5, 0: kept the candidate $x = 0$ that the squaring step introduced*
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1. Solve $\sqrt{x+5} = 4$.

Solution: The radical is already alone, so square both sides: $x+5 = 16$, hence $x = 11$.
Check: $\sqrt{11+5} = \sqrt{16} = 4$ ✓

$x = 11$

2. Fall time $t = 0.45\sqrt{d}$ seconds, d in metres, $t = 2.7$ s. Find the drop height.

Solution: Isolate the radical, then square:

$$0.45\sqrt{d} = 2.7$$

$$\sqrt{d} = 6$$

$$d = 36.$$

Check: $0.45\sqrt{36} = 0.45 \times 6 = 2.7 \checkmark$

$d = 36 \text{ m}$

3. $\sqrt{2x+3} = x$. (a) Solve. (b) The left side at $x = -1$.

Solution (a): Square both sides: $2x+3 = x^2$, so $x^2 - 2x - 3 = 0$ and $(x-3)(x+1) = 0$. The candidates are $x = 3$ and $x = -1$. Testing in the *original* equation: $\sqrt{9} = 3 \checkmark$, while $\sqrt{1} = 1 \neq -1 \times$. Only one survives.

$x = 3$

Solution (b): $\sqrt{2(-1)+3} = \sqrt{1}$.

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The left side is 1 and the right side is -1 : a square root cannot be negative, which is why the squaring step manufactured this candidate.

4. Solve $2\sqrt{x-1} + 3 = 11$.

Solution: Isolate the radical before squaring: $2\sqrt{x-1} = 8$, so $\sqrt{x-1} = 4$, so $x-1 = 16$.
Check: $2\sqrt{17-1} + 3 = 2(4) + 3 = 11 \checkmark$

$x = 17$

5. Skid marks: $s = \sqrt{21d}$ mph, d in feet, $s = 42$. Find d .

Solution: Square both sides and divide:

$$\sqrt{21d} = 42$$

$$21d = 1764$$

$$d = 84.$$

Check: $\sqrt{21 \times 84} = \sqrt{1764} = 42 \checkmark$

$d = 84 \text{ ft}$

6. $\sqrt{x+7} = x-5$. (a) Solve. (b) The left side at $x = 2$. (c) Why $x = 2$ is rejected.

Solution (a): Square: $x+7 = x^2-10x+25$, so $x^2-11x+18 = 0$ and $(x-9)(x-2) = 0$. Candidates $x = 9$ and $x = 2$. Test in the original: $\sqrt{16} = 4 = 9-5 \checkmark$; and $\sqrt{9} = 3$ while $2-5 = -3 \times$.

$x = 9$

Solution (b): $\sqrt{2+7} = \sqrt{9}$.

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Solution (c): *Instructor-judged.* At $x = 2$ the left side is 3 and the right side is -3 . Squaring destroys the sign — 3^2 and $(-3)^2$ are the same number — so the squared equation is satisfied while the original is not. Since $\sqrt{}$ never returns a negative value, no x making the right side negative can be a solution. A full-credit answer names the sign mismatch, not just "the numbers did not match".

7. Solve $\sqrt{3x+1} = \sqrt{x+9}$.

Solution: Both sides are non-negative, so squaring is safe here: $3x+1 = x+9$, so $2x = 8$. Check: $\sqrt{13} = \sqrt{13}$ ✓ $x = 4$

8. Horizon distance $D = 3.57\sqrt{h}$ km, h in metres. (a) h when $D = 35.7$. (b) D when $h = 250$.

Solution (a): $3.57\sqrt{h} = 35.7$ gives $\sqrt{h} = 10$, so $h = 100$. Check: $3.57\sqrt{100} = 35.7$ ✓ $h = 100$ m

Solution (b): $D = 3.57\sqrt{250} = 3.57 \times 15.8114 \dots = 56.4467 \dots$, which rounds to 56.45 km

9. $\sqrt{x+4} + 2 = x$. (a) Solve. (b) The left side at $x = 0$.

Solution (a): Isolate first: $\sqrt{x+4} = x-2$. Square: $x+4 = x^2-4x+4$, so $x^2-5x = 0$ and $x(x-5) = 0$. Candidates $x = 0$ and $x = 5$. Test: $\sqrt{9} + 2 = 5$ ✓; $\sqrt{4} + 2 = 4 \neq 0$ ×. $x = 5$

Solution (b): $\sqrt{0+4} + 2 = 2 + 2$. 4

Since $4 \neq 0$, the candidate $x = 0$ is extraneous.

10. Challenge: $\sqrt{x+5} - \sqrt{x} = 1$. (a) Solve. (b) Check.

Solution (a): Isolate one radical and square:

$$\begin{aligned}\sqrt{x+5} &= 1 + \sqrt{x} \\ x+5 &= 1 + 2\sqrt{x} + x \\ 4 &= 2\sqrt{x} \\ \sqrt{x} &= 2 \\ x &= 4.\end{aligned}$$

$x = 4$

Solution (b): $\sqrt{4+5} - \sqrt{4} = 3 - 2$, which equals the right side of the original equation, so the candidate is genuine. 1